Surface Topography:
Metrology and
Properties

PURPOSE-LED
PUBLISHING™

TOPICAL REVIEW

Optical wafer metrology sensors for process-robust CD and overlay control in semiconductor device manufacturing

To cite this article: Arie J den Boef 2016 *Surf. Topogr.: Metrol. Prop.* **4** 023001

View the article online for updates and enhancements.

You may also like

- Virtual experiments for the assessment of data analysis and uncertainty quantification methods in scatterometry
Gertjan Kok, Marcel van Dijk, Gerd Wübbeler et al.
- Metrology and inspection required for next generation lithography
Masafumi Asano, Ryoji Yoshikawa, Takashi Hirano et al.
- Optical dimensional metrology at Physikalisch-Technische Bundesanstalt (PTB) on deep sub-wavelength nanostructured surfaces
B Bodermann, G Ehret, J Endres et al.

This content was downloaded from IP address 149.220.140.1 on 26/10/2026 at 10:47

Surface Topography: Metrology and Properties



RECEIVED
26 September 2015

REVISED
13 January 2016

ACCEPTED FOR PUBLICATION
19 January 2016

PUBLISHED
17 February 2016

TOPICAL REVIEW

Optical wafer metrology sensors for process-robust CD and overlay control in semiconductor device manufacturing

Arie J den Boef

ASML Netherlands B.V., De Run 6501, 5504 DR Veldhoven, The Netherlands

E-mail: arie.den.boef@asml.com

Keywords: semiconductor, metrology, scatterometry, overlay, alignment, critical dimension

Abstract

This paper presents three optical wafer metrology sensors that are used in lithography for robustly measuring the shape and position of wafers and device patterns on these wafers. The first two sensors are a level sensor and an alignment sensor that measure, respectively, a wafer height map and a wafer position before a new pattern is printed on the wafer. The third sensor is an optical scatterometer that measures critical dimension-variations and overlay after the resist has been exposed and developed. These sensors have different optical concepts but they share the same challenge that sub-nm precision is required at high throughput on a large variety of processed wafers and in the presence of unknown wafer processing variations. It is the purpose of this paper to explain these challenges in more detail and give an overview of the various solutions that have been introduced over the years to come to process-robust optical wafer metrology.

Introduction

The rapid developments in the micro-electronics industry continue to have a big impact on our daily life. This is for example clearly visible in mobile communication: thirty years ago a portable telephone had a weight of about 1 kg, would cost several thousands of dollars and it only allowed phone calls. Today, a high-end smartphone weighs only about 120 g costs a few hundred dollars, fits in your pocket and offers a wealth of functions that were unheard of 30 years ago like navigation, high-speed global connectivity and built-in camera's with high-definition video capabilities.

These rapid developments have been enabled by a continuous and aggressive reduction of the dimensions of semiconductor devices according to the well-known 'Moore's law' [1]. Smaller device features help to reduce power consumption, increase processing speed and allow more functionality in a smaller volume. Moreover, the high productivity of today's semiconductor industry helps to dramatically reduce cost which makes semiconductor devices cost-effective for many applications and affordable for a large population.

Figure 1 shows a simplified drawing of the various repetitive process steps that take place during semiconductor manufacturing [2]. The patterning of the device features takes place during optical lithography where a mask ('reticle') is imaged with a 4× reduction factor on a resist-coated wafer with a diameter of 300 mm. The field size of the image on the wafer is about 25 × 36 mm^2 and the lithography tool prints this field many times on the wafer in a step-and-scan fashion. After the lithography step the resist is developed and the wafer undergoes further processing steps like etching or ion implantation.

Improvements in optical lithography have been an important driver in the continuous reduction of semiconductor device features. Over time, the exposure wavelength has been reduced from UV wavelengths down to extreme ultraviolet (EUV) wavelengths (13.5 nm) while the numerical aperture (NA) of the imaging optics has increased to a hyper-NA of 1.35 for high-end deep-UV (DUV) systems [3] and 0.33 for the latest EUV systems [4]. As a result of this continuous progress in lithography, devices with a minimum feature size or critical dimension (CD) as small as 10 nm are already made using EUV lithography or multi-patterning techniques using DUV hyper-NA lithography tools [5]. These small features must also be printed accurately on top of the underlying patterns. A misalignment results in a so-called overlay (OV) error that can result in a non-yielding device. Today the

© 2016 IOP Publishing Ltd

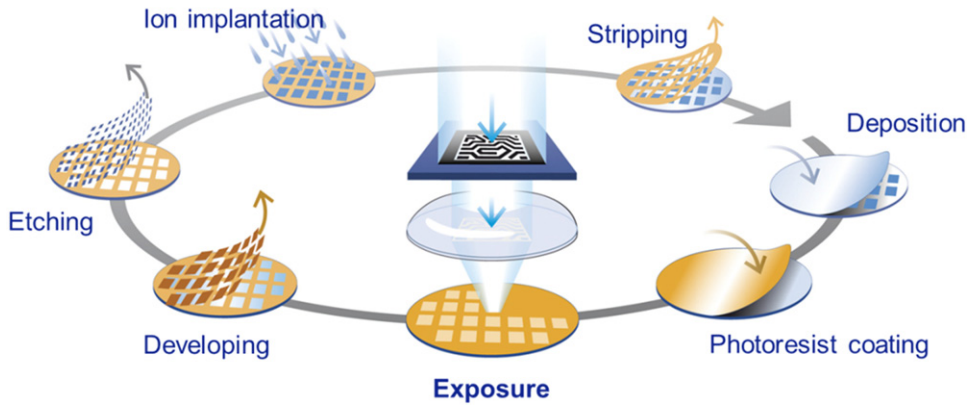


Figure 1. Simplified drawing of the repetitive production process of a semiconductor device. The lithography step where the patterning of a new product layer start is marked with 'exposure'.

maximum allowed OV error is as low as 4 nm for critical layers. This requirement is defined as the 'mean + 3σ OV variation' and needs to be realized over the entire area of a 300 mm wafer which also raises significant mechatronic challenges in a lithography tool. The OV requirement will also get tighter over time as device features continue to shrink and according to the International Technology Roadmap for Semiconductors (www.itrs2.net) it will approach the 2 nm level somewhere in 2018.

For good device performance we not only need sufficiently low OV numbers but we also need a uniform CD across the wafer. This is called 'CD-uniformity' (CDU) and as a rule of thumb, the 3-sigma variation of the CD across the wafer as measured with a CD-metrology tool like an electron microscope or an optical scatterometry tool should be less than ≈10% of the nominal CD.

Printing these small patterns raises significant challenges for the wafer metrology systems that are being used in lithography. For example, using a high NA and short wavelength yields a small depth-of-focus (DOF) which requires an optical level sensor that can measure wafer height variations with nm accuracy. Moreover, these small patterns must be printed with nm placement accuracy relative to underlying product patterns that are already present. This requires measurements of the position of the wafer with sub-nm precision which is achieved with an optical alignment sensor. After the resist has been exposed with a new pattern, the resist is developed and the resulting resist pattern is transferred into a product layer via an etch process step.

Over the years the patterning process (lithography + etch) has been pushed close to its limits. As a result the impact of small variations in the manufacturing process continues to increase to a level where small process excursions lead to significant variations in CDU and OV. Without regular monitoring and control of these parameters significant yield losses occur that are minimized by metrology of CDU and OV on the patterned wafer.

This paper presents a more detailed description of the three above-mentioned sensors: a level sensor, an alignment sensor and a metrology sensor for measuring OV and CDU. The big challenge that these sensors have in common is the need for robust sub-nm measurement precision at high productivity (>200 wafers per hour) on processed wafers with an unknown and varying stack of product layers. Moreover, the alignment and level sensor need to be mounted in a mechanically stable environment that is also carefully maintained at a stable temperature. For each sensor we will briefly describe possible measurement concepts and explain the preferred concept and show how this has evolved towards a sensor as it is being used today.

The level sensor

In lithography, a mask pattern is projected on a resist film using high-NA imaging optics. Small focus offsets will lead to CD variations which degrades CDU. With today's wavelengths and NA the DOF is roughly of the order of 50 nm which is much smaller than the thickness variations of a silicon wafer. Moreover, the wafer is covered with processed thin films that can have considerable thickness variations across the wafer. As a result, lithography needs a so-called level sensor that measures the resist height variation across the wafer before it is exposed. This measurement must be done with nm-precision and it must be able to map the full wafer surface of every wafer at measurement speeds as high as ≈240 wafers per hour. In most cases wafer-height varies gradually across the wafer so a spatial resolution of about $1 \times 1\,\mathrm{mm}^2$ is normally sufficient.

A mechanical profilometer [6] has the required height and spatial resolution but a single profilometer stylus is too slow to measure a full wafer height map in

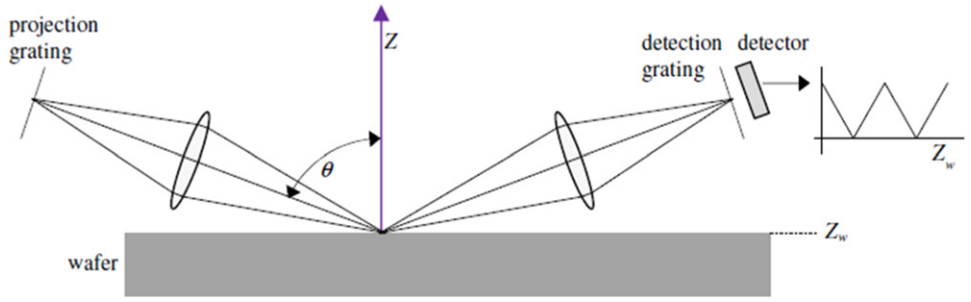


Figure 2. Conceptual drawing of the level sensor. A projection grating is projected on the wafer surface and the reflected image is re-imaged on a detection grating. This image will shift laterally if the wafer height varies resulting in a variation of the intensity that is transmitted by the detection grating and measured by the photo detector.

the available time. Moreover, the direct mechanical contact with the wafer surface can lead to wafer defects and contamination which is unacceptable for reasons of production yield.

Acoustic techniques [7] are non-contact, have sufficient spatial resolution and the measurement speed can be made sufficiently high by using parallelism. However, the acoustic wavelength is too long compared to the required resolution. Acoustic techniques are therefore not usable for high-end semiconductor device manufacturing. Moreover, acoustic height sensing techniques cannot be easily used in a vacuum environment that exists inside an EUV lithography tool.

Capacitive height sensors [6] have been used in earlier lithography tools. These sensors meet requirements like resolution and measurement speed. Moreover it is also a non-contact technique and it can operate in vacuum. However, the response of a capacitive sensor is highly dependent on the electrical properties of the substrate. As a result, large height measurement errors can occur that depend on the properties of the stack of thin films that are present on the wafer. In principle, these errors can be calibrated with a send-ahead wafer but there are always wafer-to-wafer film thickness variations that lead to uncorrectable measurement errors that exceed the DOF requirements.

Optical sensor concepts are another obvious possibility: they are non-contact and offer sufficient measurement speed, precision and spatial resolution. Interferometry is one possible optical concept but it requires a stable reference beam. The need for a stable reference beam is avoided with a differential interferometer that measures the local gradient of a surface [8]. A height map of the surface is then obtained by integrating the measured gradient. In this approach, however, small measurement errors and low-frequency noise lead to height errors that can exceed the available DOF. For these reasons, optical interferometric concepts are not commonly used for wafer height mapping in lithography.

Optical triangulation has none of the above challenges. It does not require a stable reference beam and it can measure a wafer height with sufficient speed, resolution and precision which makes it an ideal choice for a level sensor in lithography [9, 10].

The concept of optical triangulation is simple and is schematically shown in figure 2.

An amplitude grating ('projection grating') is imaged on the wafer surface at an angle θ relative to the surface normal. The grating image is reflected by the wafer surface and re-imaged on an identical grating ('detection grating'). Due to the oblique incidence, a wafer height variation Z_w will shift the image of the projection grating on the detection grating over a distance s given by [9]:

$$s = 2Z_w \sin(\theta).$$

The shifted image of the projection grating is partially transmitted by a detection grating and the transmitted intensity, which is a periodic function of the image shift, is measured with a photodiode. The size of the grating determines the spatial resolution and is sufficiently large to average-out any surface roughness that may be present on the substrate. The pitch of the grating and the angle-of-incidence determine the height resolution. By picking a sufficiently large angle-of-incidence and a sufficiently small pitch wafer height variations as low as 1 nm can be detected.

However, process layers on the silicon substrate pose an extra challenge. Light is partially transmitted by these layers as shown in figure 3 and results in a measurement offset called apparent surface depression (ASD). It can be shown with a plane-wave description of the triangulation concept that the measured height (Z_m) is given by [9]:

$$Z_m = \frac{\lambda}{4\pi \sin(\theta)} \frac{\partial \psi}{\partial \theta}$$

where ψ is the phase of a reflected plane wave at an incident angle θ and λ is the wavelength of the incident light.

This measurement offset is further clarified in the left-hand drawing of figure 3 which shows how an incident plane wave is reflected at various interfaces.

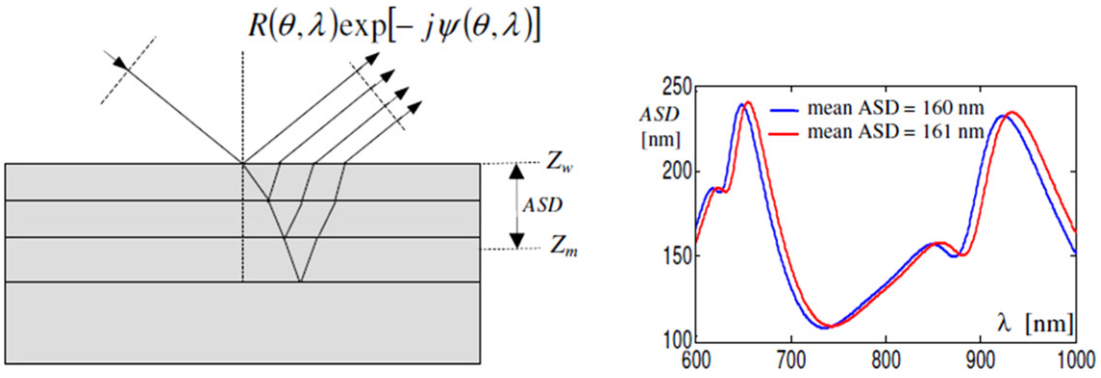


Figure 3. The left-hand drawing shows a plane wave propagating into a thin-film stack where it is partially reflected at the interfaces. As a result of these multiple reflections, the measured height (Z_m) is not equal to the actual height (Z_w) of the resist film. This difference is called apparent surface depression (ASD) and is wavelength-dependent as shown in the right-hand graph for an oxide film on a silicon wafer. The blue curve has been calculated for a 1 μm oxide thickness and the red curve has been calculated for an oxide thickness of 1.01 μm. The ASD as averaged over all wavelengths ('mean ASD') varies only 1 nm for this 10 nm oxide thickness variation.

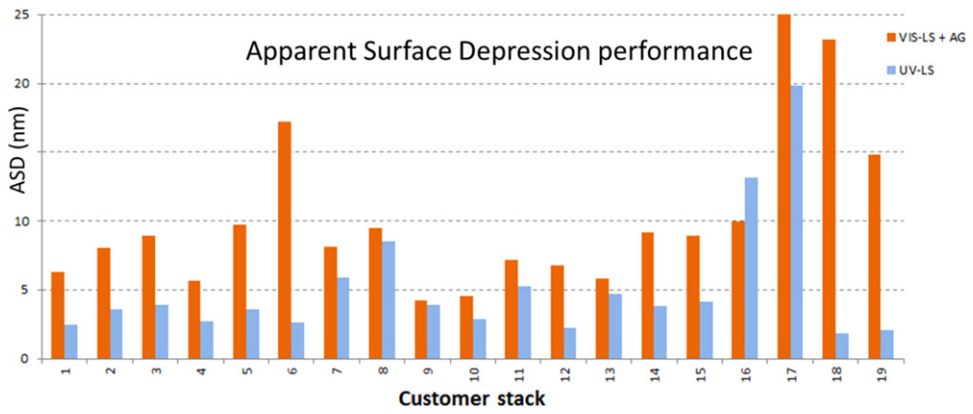


Figure 4. Measured ASD for different stacks. The orange bars shows the ASD for a level sensor using visible light and air gauge ('VIS − LS + AG') and the blue bars show the measured performance for a level sensor using UV light and no air gauge ('UV-LS').

All reflected waves add-up coherently resulting in a wavelength-dependent phase ψ of the reflected field. The right-hand graph of figure 3 shows a calculated ASD as a function of wavelength for an oxide film on a silicon substrate for 1000 nm thickness (blue curve) and 1010 nm thickness (red curve). It can be seen that certain wavelengths are sensitive to small film thickness variations while other wavelengths turn out to be very robust. For example, around $\lambda = 680$ nm the small thickness variation causes a very large ASD variation while the ASD is very stable for $\lambda \approx 860$ nm. For this reason, level sensors in lithography use broadband light: In this way the large wavelength-dependent measurement errors are averaged-out resulting in a more stable wafer height measurement over a wide range of stack variations. In the example as shown above, a 10 nm oxide thickness variation gives only a 1 nm variation in measured wafer height which is sufficient to realize the tight focus requirements of today. This residual focus error is much smaller than the 50 nm DOF requirement but one must keep in mind that there are more potential error sources like wafer unflatness that will also contribute to the total focus error.

In order to further drive down any residual focus error, lithography tools are optionally equipped with an additional non-optical sensor that is based on measuring airflow between a nozzle and the wafer surface [11]. Height variations between the wafer surface and the nozzle will change the airflow and these changes are detected with a small built-in flow sensor. The bandwidth of such an air flow sensor, however, is too small for measuring every wafer. It is therefore only used to calibrate and correct the residual mean ASD.

Recently, a further significant improvement of the process-robustness of the level sensor has been achieved by replacing the visible light source by broadband UV light [12]. Many process layers and materials like amorphous carbon and poly silicon that are used in semiconductor manufacturing absorb UV light. As a result, the UV light that is used in the level sensor is partly reflected from the resist film and strongly

absorbed in underlying films. In this way, deeper lying product structures are almost invisible to the level sensor light which reduces the ASD error. Figure 4 shows how the use of UV light reduces the ASD reduction on different customer stacks. It can be seen that the ASD shows large stack-to-stack variations which is due to the different composition of process layers for each different stack. Moreover, it can also be seen that the improvement that is obtained by going to UV light shows a very large variability. On some stacks (like 6, 18 and 19) the improvement is very significant but on a few stacks the improvement is small (like 8 and 11) or even absent (stack 16). This is caused by the optical properties of the process layers in the customer stacks. Some layers are transparent at visible wavelengths and become completely opaque in the UV range leading to a large improvement. Other layers like used in stacks 8 and 11 are already relatively opaque in the visible range. In that case moving towards the UV will not give a big improvement. . Further details and measured improvements on actual product wafers are reported in [12].

The alignment sensor

Inside a lithography tool a wafer is placed on a wafer table with a robot gripper. The repeatability of this placement is of the order of several microns so in order to print the new resist pattern on top of underlying product patterns the exact lateral position of the wafer must be measured with an alignment sensor before the wafer can be exposed. Unfortunately, wafer processing makes the measurement of a wafer position more challenging than expected. Wafer processing steps like rapid thermal anneal (RTA) can cause large in-plane wafer distortions. The presence of many process films on a silicon wafer introduces mechanical stress that causes significant wafer warp. Clamping such a warped wafer on a flat wafer table also results in significant in-plane wafer distortions.

As a result, wafer position cannot be defined by only three numbers (X, Y and rotation). Due to the in-plane wafer distortion the alignment sensor has to measure a wafer location on many (≈20–40) points on a wafer to allow an accurate determination of the deformed wafer grid. To allow this position measurement one or more so-called alignment targets are printed in every field on the wafer. These targets have dimensions of about $40 \times 40\ \mu m^2$ and they are printed in the scribe lane area between the individual product dies [13].

For a unique position determination a target must be 180° rotation-symmetric which ensures that an optical sensor will measure the target's center-of-gravity as the target position. Many target designs are possible but for highest precision gratings are commonly used since the periodic nature of these targets allows efficient noise suppression by spatial filtering.

A small grating pitch helps to improve precision. Ideally one would like to use gratings with a pitch that is equal to the pitch of the device patterns but unfortunately, there is a lower limit to the pitch that is imposed by the shortest possible measurement wavelength. Wavelengths in the UV range are not possible due to the risk of resist exposure. Moreover, alignment gratings are often covered by materials like polysilicon and amorphous carbon that are not transparent at wavelengths in the UV range. In order to ensure sufficiently good measurement quality the pitch of an alignment grating is often in the range of 1–5 μm [13].

One particular challenge that any alignment sensor concept must be able to deal with is the fact that alignment gratings can have a very low optical contrast due to absorption of light in process films and the need for so-called process segmentation. Diffraction efficiencies of the 1st diffraction order can sometimes become as low as $\approx 10^{-3}$%.

One possible alignment sensor concept is optical microscopy where the target is illuminated with broadband spatially incoherent light. The light that is diffracted by the alignment target is collimated by a high-quality objective and a magnified image of the grating is formed on a CCD or CMOS array detector [14, 15]. Pattern recognition is used to locate the alignment grating and image processing algorithms determine the exact location of the grating relative to a sensor-internal reference target that is also projected on the detector array. This method is conceptually simple but it has a few challenges:

(1) On weak gratings with diffraction efficiencies less than about 1% one could use dark-field imaging. This yields high-contrast images but with very low signal levels where read-noise of the detector array limits the performance. Bright-field imaging yields higher signal levels but at a low contrast that is also process- and focus-dependent.

(2) Lens aberrations in the sensor optics lead to small offsets in the measured target positions. These errors are of the order of 1 nm. This may seem small but one has to keep in mind that it already consumes a large part of the 4 nm overlay requirements of today. Ideally, these errors can be removed via a calibration. However these aberration-induced errors depend on the diffraction signature (diffraction efficiency as function of angle and wavelength) of the grating that changes when there is a process variation like a film thickness variation on the wafer. These variations are unavoidable in a real manufacturing environment and the resulting errors can only be minimized by illuminating the grating at a well-defined wavelength and a small illumination NA which goes at the expense of available light flux.

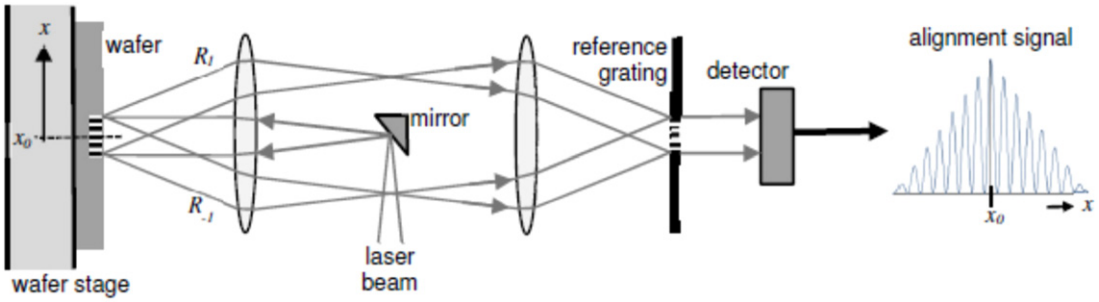


Figure 5. Conceptual sketch of a phase-grating alignment sensor. An alignment grating is illuminated with a laser beam. The +1st and −1st order are used to project a filtered image of the grating on a reference grating. The light transmitted by the reference grating is picked-up by a detector which detects the transmitted intensity while the alignment grating is being translated.

These challenges are effectively solved with a laser-based phase-grating alignment (PGA) concept which was already described in a US patent as early as 1974 [16]. In PGA a laser beam illuminates an alignment grating with a pitch P as shown in figure 5 [10]. The 0th diffracted order (i.e. the specular reflection) is blocked and a spatially filtered dark-field image of the grating is projected on a fixed reference grating using only the −1st and +1st diffraction orders. As a result of this particular form of spatial filtering, the aerial image of the alignment grating has a sinusoidal pattern with a 100% contrast and a pitch $P/2$. The total transmitted intensity of the light that is transmitted by the reference grating is measured with a photodiode while the wafer is scanned at a speed V of several mm s^{-1} relative to the alignment sensor. This results in a sinusoidal signal variation with a frequency $2V/P$ and a very slowly varying envelope that is determined by the length of the gratings.

In practice, one needs to measure an X and a Y location with one sensor which is realized by putting a Y-reference grating alongside the X-reference grating. Moreover, in order to correctly determine which peak of the signal curve points to the correct location two additional X, Y gratings with slightly different pitches can be added as well as shown in figure 6. The use of two slightly different pitches of, respectively, 16 μm and 17.6 μm allows an unambiguous determination of the marker position using the nonius principle [17].

This PGA concept has some clear benefits that are particularly relevant in a semiconductor manufacturing environment:

(1) The spatial filtering and the convolution with a reference grating results in a strong suppression of process noise introduced for example by material grains of deposited films.

(2) A detector array is no longer needed for image capture. Instead, one simple 'point' detector (e.g. a PIN photodiode) per grating is needed. Such a detector offers quantum-noise-limited detection at a high measurement speed even at diffraction efficiencies as low as 0.01%.

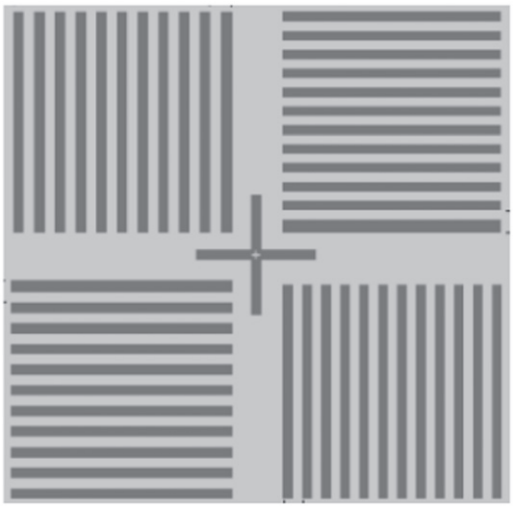


Figure 6. Example of an alignment target consisting of four gratings. The top-left and bottom-right quadrants are two X-gratings with a slightly different pitch of 16 μm and 17.6 μm, respectively. The top-right and bottom-left gratings are Y-gratings with the same slightly different pitches.

(3) PGA yields a relatively small angular divergence of the diffracted beams at a well-defined wavelength. As a result, the distribution of the diffracted light in angle and wavelength is stable in the presence of processing variations which results in negligibly small uncorrectable sensor-induced offsets.

The performance of the PGA concept is illustrated in figure 7. The left-hand drawing shows that PGA is able to achieve sub-nm Overlay performance on a batch of 300 mm etched bare silicon test wafers using a wavelength of 633 nm (HeNe laser). The right-hand drawing demonstrates that the alignment repro stays well-below 1 nm even if the diffracted light is attenuated by more than three orders of magnitude by inserting neutral density absorption filters in the illumination beam.

These performance results have resulted in a widespread acceptance of the PGA concept in semiconductor device manufacturing. PGA was first

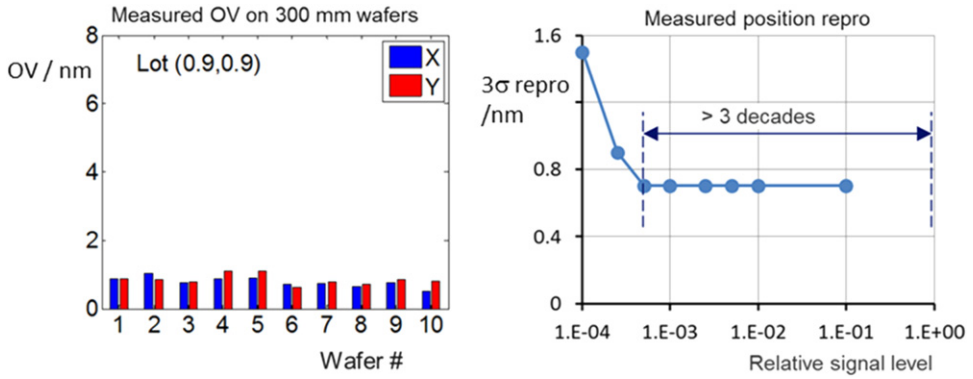


Figure 7. Measured performance of a phase grating alignment (PGA) system on etched bare silicon test wafers. The left-hand graph shows that PGA can offer sub-nm overlay performance on a set of wafers and the right-hand graph shows the measured alignment reproducibility of an alignment measurement as a function of the signal strength.

reported in [17] and was introduced as a through-the-lens (TTL) alignment sensor around 1981. In TTL the grating pitch was 16 μm and it used a HeNe laser as a light source. TTL has been very successfully used for many years but due to the progression of the semiconductor manufacturing technology, TTL started to face some challenges on new processes:

(1) On some layers the optical depth of an alignment grating could become equal to $\lambda/2$. In that case the 1st order vanishes resulting in a complete loss of signal.

(2) Tighter focus requirements forced chip manufacturers to planarize processed wafers with chemical mechanical polishing (CMP). CMP, however, is optimized for sub-micron device patterns. Coarse alignment gratings with a 16 μm pitch were asymmetrically distorted by CMP resulting in large alignment offsets.

In order to address these challenges the PGA alignment sensor concept was extended with a 2nd wavelength (532 nm) and the capability to align on gratings with smaller pitches given by 16 μm/N ($N \in [1\ldots7]$). This extra wavelength and the ability to have some pitch flexibility proved to be a powerful extension of the PGA concept that reduced the impact of asymmetric grating deformation and which prevented the loss of alignment signal for marker depths equal to $\lambda/2\ldots$ However, as devices continued to shrink new processes and new materials were introduced resulting in additional challenges:

(1) Hardmask materials like amorphous carbon were introduced for improved pattern transfer of the resist pattern into the underlying product films. These materials absorb visible light and only become transparent for near-IR light.

(2) There was a growing need for even more freedom in optimizing the alignment grating pitch and the grating orientation. Simple X and Y gratings were not always optimal and there was a need to look into gratings with arbitrary pitches and orientations (like 45° relative to the X-axis).

The first challenge is relatively easy to solve by adding two more near-IR wavelengths (780 and 850 nm). An optical multiplexer can be used to merge the four laser beams into one illumination beam and an optical de-multiplexer can be used to separate and guide the four wavelengths to four detectors. The 2nd challenge is more difficult to tackle since this cannot be easily solved adding more reference gratings in the sensor.

However, it turns out that this challenge can be elegantly solved with a compact self-referencing interferometer using rotation prisms [13]. Before explaining this solution in more detail it is helpful to point out that a PGA concept is basically a two-beam interferometer: the two intersecting beams of the −1st and +1st order form interference fringes on a reference grating. As the grating is slowly scanned the phases of these orders will move in opposite directions resulting in a translation of the fringes that is measured with the reference grating. In other words, PGA essentially measures the phase difference between the −1st and +1st orders. With this in mind it is now easier to explain the use of a self-referencing interferometer in PGA as shown in figures 8 and 9.

The interferometer splits the diffracted orders in two identical parts that are rotated 180° relative to each other and recombined. The resulting output of the interferometer is a perfect overlap of opposite diffraction orders that produces an interference signal for any order-pair (n, $-n$) and any gratings orientation. As a result only one detector per wavelength is needed for alignment using a grating that can have any pitch and any orientation [18].

This self-referencing concept is now being used in the alignment sensors for high-end semiconductor manufacturing. The availability of four wavelengths enables a robust alignment on virtually any stack of

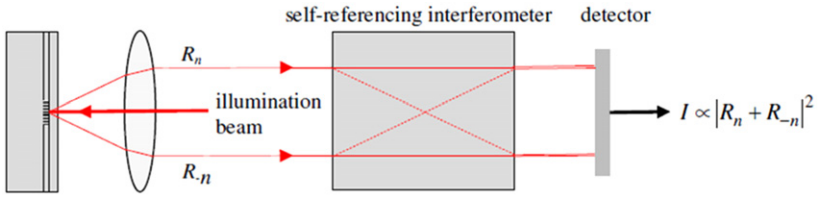


Figure 8. Conceptual drawing of an alignment sensor based on a self-reference interferometer.

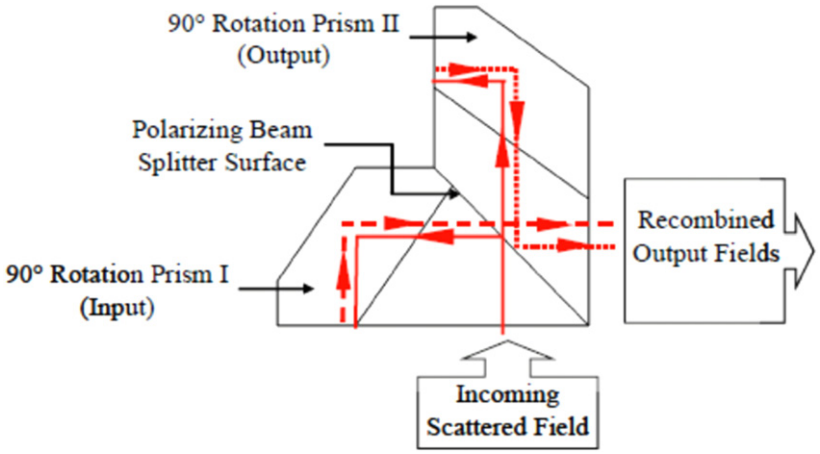


Figure 9. A more detailed drawing the of the self-referencing interferometer. The incoming beam is split in two beams. One beam is rotated over +90° and the other beam is rotated over −90°. The two rotated beams are merged with a beam splitter resulting in overlapping beams that are rotated 180° relative to each other.

layers that is being used today. Moreover, this alignment sensor concept offers complete design freedom of the alignment grating in terms of pitch and orientation while at the same time maintaining backwards compatibility and all the benefits of the original PGA concept.

CD metrology

After lithography the resist pattern on the wafer is inspected before the wafer undergoes an irreversible process step like etching or ion implantation. One of the metrology steps is a measurement of the CD of the device patterns or, more precisely, the uniformity of the CD (CDU). Large CD variations degrade device performance. As a rule-of-thumb the CDU must be less than 10% of the nominal CD. With such a tight requirement variations in CD must be detected as early as possible with an in-line CD metrology tool so that CD changes can be corrected via small adjustments of the exposure dose or focus inside the lithography tool. An excellent and comprehensive overview of CD metrology requirements, challenges and potential measurement techniques has already been presented in [19] so we will limit ourselves here to a brief summary of various techniques that may be considered for CD control applications.

In the early days of the semiconductor industry device patterns were so large that optical microscopy was used for CD metrology [20]. However, because of the aggressive shrink of device features this is no longer feasible. Today devices with a CD of only 10 nm are being developed and even DUV immersion microscopes cannot resolve dense patterns of 10 nm features. As a result electron microscopy ('CDSEM') is now commonly used for metrology of the CD of device patterns [21]. CDSEM has sufficient resolution but also has some drawbacks:

(1) The relatively high energy of the electrons (≈800 eV) results in shrink of the resist lines [22].

(2) The noise levels are relatively high compared to today's requirements resulting in long averaging times which reduces throughput.

AFM can also be used for CD metrology [23] but the low measurement speed and the tip lifetime makes it a less obvious choice for high-speed in-line CD metrology.

Small angle x-ray scattering (SAXS) is another high-resolution technique that has proven to be capable for metrology of CD [24–26]. The sub-nm wavelength of available x-ray sources certainly provides sufficient resolution for metrology of relevant patterning parameters like CD and linewidth roughness. Unfortunately, the brightness of existing tabletop x-ray sources is still insufficient to reach (sub)-second

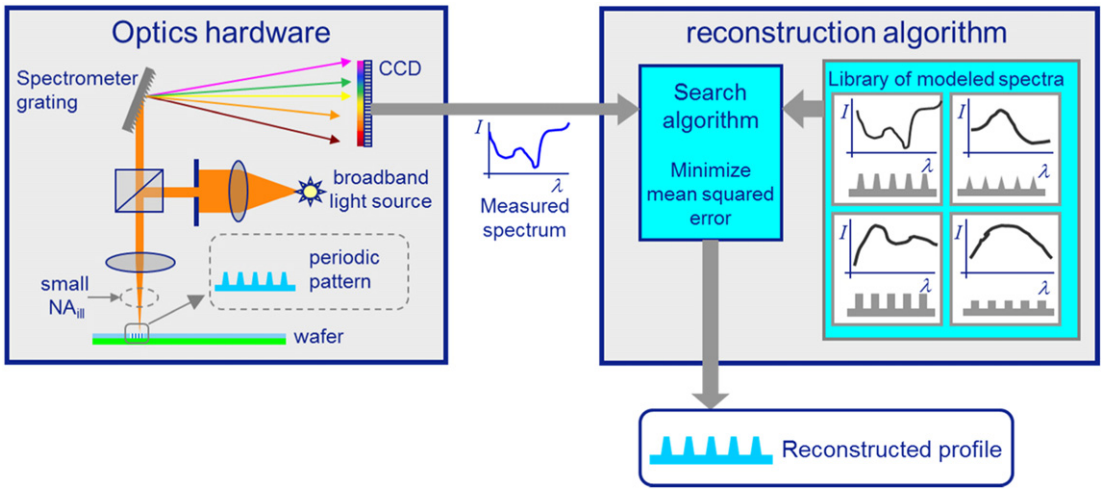


Figure 10. Conceptual drawing of a spectroscopic scatterometer for CD metrology. A periodic pattern is illuminated with broadband light and the spectrum of the reflected light is measured. In order to reconstruct the profile of the periodic pattern, the measured spectrum is compared with a set of pre-calculated spectra that are stored in a library.

acquisition times on small ($\approx 20 \times 20\ \mu m^2$) metrology targets. This makes SAXS less suitable for in-line CD metrology applications and its applications space is more focused on reference metrology.

Fortunately, over the years optical scatterometry [27] has gradually evolved into a suitable in-line CD metrology technique (often referred to as optical CD: OCD) that offers sub-nm precision at measurement times as low as ≈0.1 s per sample. There is a vast amount of literature available on OCD for example [27–30] so here we limit ourselves to a summary of the operating principle.

A highly simplified spectroscopic scatterometry concept is shown in figure 10. A periodic pattern is illuminated with broadband light at a well-defined angle-of-incidence. The specular reflection is collimated and its intensity is measured as a function of wavelength. The measured spectrum depends on the shape of the periodic pattern. If the shape is known we can calculate the spectrum (forward problem) but in this case we are dealing with an inverse scattering problem: We know the spectrum and we need to find the shape parameters like height and width (=CD) of the grating.

This reconstruction problem can be solved by first creating a library that contains pre-calculated spectra for different grating profiles. A library search is then done to find the modeled spectrum that yields a best match with the measured spectrum. Once the best match is found we also have the shape of the grating. For simplicity, figure 9 shows only four library entries but it should be emphasized that a real library for more complex structures can contain thousands of entries.

This concept looks simple but it has a very large process-robustness challenge. In a real CD metrology application the resist pattern is always on top of a stack of thin films of different materials. Just like in the level sensor, the illumination beam is partially transmitted by these layers which make the measured spectrum also dependent on the properties (thickness and complex refractive index) of the film stack. Often the film thickness varies during the manufacturing process so for a robust decoupling of CD variations from film thickness variations we need to include film thickness in the reconstruction. This requires measuring more independent information which is done by polarizing the illumination beam and measuring the intensity ('reflectometry') or polarization state ('ellipsometry') of the reflected light for different input polarizations. This significantly improves the robustness of OCD but a validation of the reconstructed CD with a CD-SEM is often still done before the OCD results are used for monitoring CD variations in a high-volume production environment.

Today spectroscopic scatterometry is the most common OCD tool but it has a few challenges:

(1) The illumination NA must be kept small to minimize smearing of spectra from different incident angles. The reconstruction SW can take this smearing into account but in practice the NA is limited to about 0.1. In order to be competitive with a CD-SEM there is also a need to measure on small gratings less than $10 \times 10\ \mu m^2$ which requires a small spot size with a diameter of only a few μm. The combination of a small spot size, a small illumination NA and an extended incoherent source leads to very low light levels which makes it challenging to reach sub-second acquisition times on small gratings.

(2) A spectroscopic OCD tool only measures the 0th diffraction order[1]. In the next section we will show

[1] However, in order to improve a process-robust measurement of lithography parameters, the use of the 1st diffraction order has been proposed in [38].

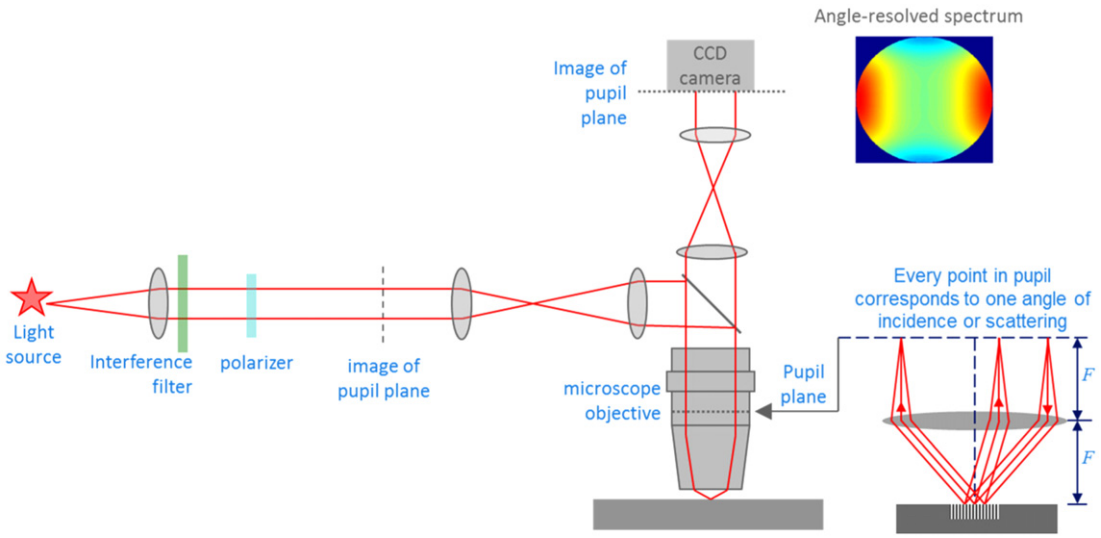


Figure 11. Conceptual drawing of an angle-resolved scatterometer. A periodic pattern is illuminated with a cone of angles with quasi-monochromatic polarized light. The diffracted light is collected with a high-NA objective forming an angle-resolved scattering spectrum of the target in the pupil plane of the objective. A simple piece of relay optics is used to project this angle-resolved spectrum on a CCD array.

that the ability to also measure the +1st and −1st diffraction orders enables process-robust OV metrology which is also of considerable interest in lithography.

Angle-resolved scatterometry shown in figure 11 is an alternative concept that addresses these challenges [29]. It relies on the basic lens property that every point in the back-focal-plane corresponds to one angular direction in the image plane. A light source illuminates the periodic pattern via a high-NA microscope objective with polarized quasi-monochromatic light. The large NA ensures that the grating is illuminated over a large range of incident angles from 0° up to ≈72° which results in a large 'information content' in the light that is diffracted by the grating. This diffracted light is collimated by the objective and forms an angle-resolved scattering spectrum in the back-focal plane. In most objectives the back-focal plane is localized inside the objective's housing so some simple relay optics is used to re-image the back-focal plane on a detector array where the angle-resolved spectrum is captured and used for reconstructing the grating and underlying thin films.

This concept is complementary to spectroscopic scatterometry. Instead of measuring the diffracted light as function of wavelength at a fixed angle, we now measure as a function of angle (in two dimensions) at an adjustable wavelength. This approach has some benefits that are particularly relevant in the area of CD and overlay metrology:

(1) The high NA yields a large photon collection efficiency. Even at small spot sizes short acquisition times well below 1 s are possible.

(2) The high NA also results in small illumination spots that allow measurements on gratings smaller than 10 × 10 μm^2.

(3) This concept can easily measure higher diffraction orders. In the next section we will show how this can be used to measure overlay as well which allows us to use one single metrology tool for both overlay and CD metrology.

Figure 12 shows a measured CD result that was obtained on an array of contact holes on top of a thin film stack [31]. In this particular case one of the layers was a birefringent amorphous carbon layer [32]. The optical properties of this layer were varying across the wafer but despite this process variation, angle-resolved scatterometry was able to reconstruct the variation of the refractive index [31] and obtain a measured CD result that correlated very well with a CDSEM.

Every dot in the graph represents a measured CD on a 40 × 40 μm^2 contact hole array on the wafer. This wafer was exposed with a programmed exposure energy variation across the wafer. In this way a relatively large CD variation is present which allowed us to investigate the correlation with a CD-SEM [31]. The measured results in the graph show that there is indeed a good correlation between scatterometry and CD-SEM. However, a significant offset of about 12 nm is present. This is due to the fact that the image formation via electron scattering is different from the model-based image reconstruction in optical scatterometry. Moreover, the image processing that is applied to CD-SEM images also impacts the measured feature dimensions. Fortunately, once a correlation between CD-SEM and scatterometry has been established, this

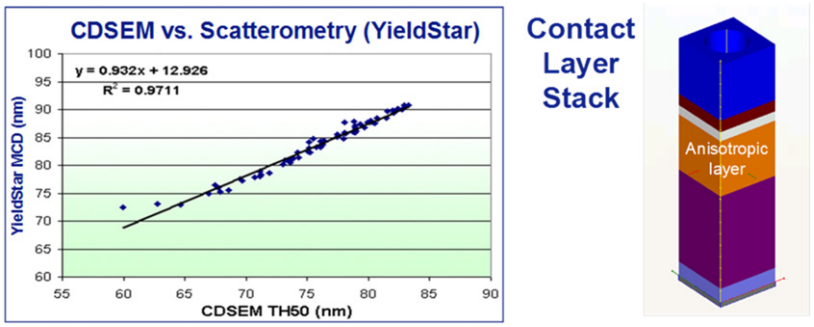


Figure 12. CD of a contact hole array as measured with an angle-resolved scatterometer. The right-hand graph shows that various thin film layers were present underneath the contact hole layer. One of those layers was a birefringent amorphous carbon film. The left-hand graph shows the measured CD variation on a 300 mm test wafer. It can be seen that the measured scatterometry result shows an excellent correlation with the results that have been obtained with a CD-SEM.

measured offset is of no concern and it can be used as a fixed correction factor in all scatterometry CD measurements.

Overlay metrology

As a rule-of-thumb, overlay should be less than about 20%–30% of the nominal CD. According to the ITRS roadmap this would correspond to an OV requirement of 4 nm at today's feature sizes. This overlay requirement has to be realized on a wafer with a 300 mm diameter which is a very challenging requirement that can be impacted by many possible error sources. Even a highly accurate alignment of the wafer prior to exposure is not yet a guarantee for perfect OV. Various effects can still contribute to an OV error . For example:

(1) Alignment measures the location of the wafer on the wafer stage. This does not yet guarantee that the new image is printed exactly on top of the product patterns that are already present on the wafer. Small wafer stage positioning offsets or small projection lens aberrations lead to OV errors that are only captured during OV metrology.

(2) Wafer alignment results are used to model the in-plane wafer deformation in a 2D wafer grid. Due to the finite sampling of the alignment gratings on the wafer a small residual offset between the modeled wafer grid and the actual wafer grid can remain which leads to a small OV error.

These errors can be reduced by modeling the measured OV data as a set of correction factors that are fed back to the lithography tool. The quality of this correction, however, is completely determined by the quality of the OV metrology tool.

Ideally, one would like to measure the OV directly on the device patterns. However, as noted in the previous section, optical methods cannot resolve these structures and these kind of measurements can only be done with a CDSEM (provided that the electrons have sufficient penetration depth) or with a cross-section TEM. The latter method, however, is obviously destructive and is only used for calibrating optical metrology methods.

Just like the alignment sensor an optical OV metrology sensor requires special targets with patterns that are sufficiently coarse to be resolved with visible and near-IR light. OV metrology, however, must also be able to measure inside an exposed field (intrafield) where the product pattern is also located. Within this exposure field there is almost no room available for metrology targets so OV targets have to be significantly smaller than alignment gratings. In practice the size of an OV target is about $10 \times 10\,\mu m^2$ and several of these targets are distributed within each exposed field. Today there is also a clear trend towards using gratings for OV metrology for reasons that are similar as alignment.

There are two optical OV metrology concepts: image based OV metrology (IBO) and diffraction based OV metrology (DBO). In case of IBO an OV target is built up of X and Y resist gratings that are spatially separated from X and Y gratings in the product layer. An example of such a grating with 180° rotation symmetry is shown in figure 13.

A high-quality microscope images this target on a detector array and pattern recognition plus image processing determines the average distance of the resist gratings from the average location of the product gratings. In IBO lens aberrations and illumination imperfections introduce an offset called tool induced shift (TIS) that can be corrected by measuring the OV for 0° and 180° wafer orientation [33]. Unfortunately, TIS in IBO is sensitive to wafer processing variations. For example, film thickness changes will change the spectral shape and angular distribution of the light that is diffracted by the gratings which results in varying TIS numbers. This variation can only be minimized by illuminating the target at a well-defined wavelength and a well-defined illumination angle which goes at

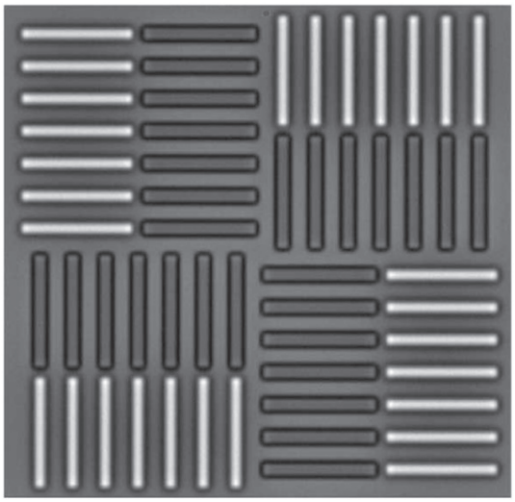


Figure 13. Example of an overlay target that is used for image based overlay metrology (IBO). The dark and light gratings are present in the different layers. The X and Y gratings are arranged in such a way that the target has 180° rotation symmetry.

the expense of signal level (resulting in long measurement times).

Many of the challenges of IBO are addressed with a different method called DBO [34]. This method also uses gratings in the top and bottom layer but in contrast to IBO, the gratings are not placed alongside each other but on top of each other. If the gratings are perfectly aligned (=zero overlay error) they form a symmetric composite grating with symmetric scattering properties. However, a small misalignment (overlay error ≠0) creates an asymmetric composite grating which creates an asymmetry in the intensity of diffracted light as schematically shown in figure 14.

This intensity asymmetry is particularly pronounced in the higher diffraction orders and for small OV errors it can be approximated by [35]:

$$\Delta I = I_1 - I_{-1} = K \times \mathrm{OV}.$$

K is a proportionality constant that depends on the measurement settings (wavelength, polarization and angle-of-incidence) and the geometry of the gratings and properties of the thin films between the gratings ('the stack'). The measurement settings are usually known quite accurately but the stack is usually unknown which makes it difficult to derive the overlay from a measured intensity asymmetry ΔI.

Fortunately, we can eliminate this K-factor in a simple and very robust way by measuring two resist gratings on top of two corresponding product gratings. For each resist grating we deliberately introduce a small lateral shift ('bias') d which results in two measured asymmetries:

$$\Delta I_+ = K(\mathrm{OV} + d),$$
$$\Delta I_- = K(\mathrm{OV} - d).$$

We have now two measured asymmetries and two unknowns (K and OV) which enable us to eliminate K and solve OV [35]:

$$\mathrm{OV} = d\frac{\Delta I_+ + \Delta I_-}{\Delta I_+ - \Delta I_-}.$$

One of the benefits of measuring overlay in this way is that it is based on an intensity measurement. Such a measurement is less susceptible to aberrations in the metrology optics than IBO which results in lower measurement errors.

An angle-resolved scatterometer can easily detect higher diffraction orders with, for example, an annular illumination mode as shown in figure 15 which makes it ideally suited for this DBO technique.

The picture of the angle resolved spectrum in figure 14 was measured on an overlay grating with a pitch of 650 nm and a wavelength of 700 nm. For these settings, the annular illumination results in a +1st and −1st diffraction orders that are clearly visible as annular segments. Moreover, one can also see a clear intensity asymmetry due to a misalignment between top and bottom grating.

As stated earlier, one of the benefits of this method is that it is less sensitive to sensor imperfections than the IBO technique which results in an improved measurement precision. This is clearly demonstrated in figure 15 which shows the so-called total measurement uncertainty (TMU) number of IBO and DBO as measured on various product layers. TMU is a figure of merit with various definitions. For OV metrology we use the following definition for TMU [36]:

$$\mathrm{TMU} = \sqrt{(3\sigma_{\mathrm{repro}})^2 + \mathrm{TIS}_{\mathrm{mean}}^2 + (3\sigma_{\mathrm{TIS}})^2}.$$

Here σ_{repro} is the dynamic repeatability of an OV measurement and is determined by measuring the OV on a target ten times. Between each measurement the wafer is unloaded from the tool and re-loaded in the tool. $\mathrm{TIS}_{\mathrm{mean}}$ is the mean value of the TIS that is measured on all targets on the wafer and the 1-sigma value of the across-wafer variation of this TIS value is denoted by σ_{TIS}. The definition of TMU as presented here is slightly different from the one that is given in [33]. First of all, in our definition the average TIS contribution across the wafer is included since this term is normally not corrected for by an extra 180° wafer rotation. Secondly, the term that represents the matching between metrology tools is not included since this is reported as a separate number as presented in [36].

It can be seen in figure 16 that there is quite some variation in TMU from layer to layer. This can have various reasons. For example, the bottom part of an OV target can be covered by an absorbing layer like a hardmask that attenuates the light. This would result in a weaker signal on the detector which increases the dynamic repro.

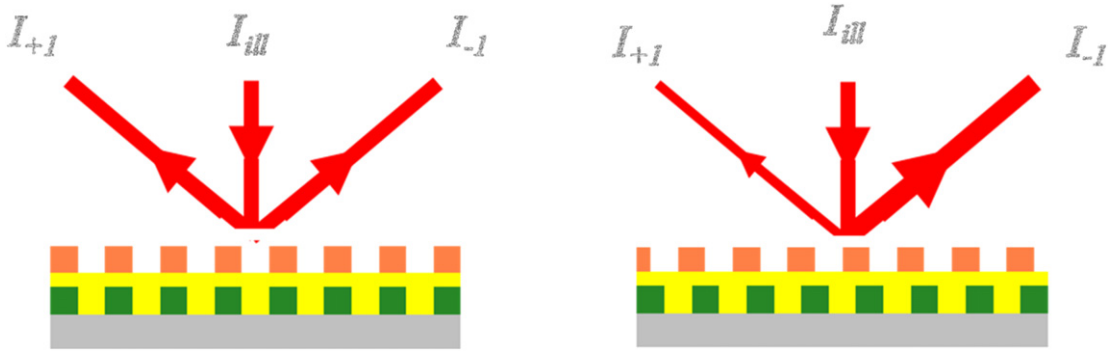


Figure 14. In diffraction based overlay metrology (DBO) the top and bottom grating are overlapping resulting in a set of optically coupled gratings. As a result of this coupling, the intensities of the diffracted orders will change when the top grating is shifted relative to the bottom grating.

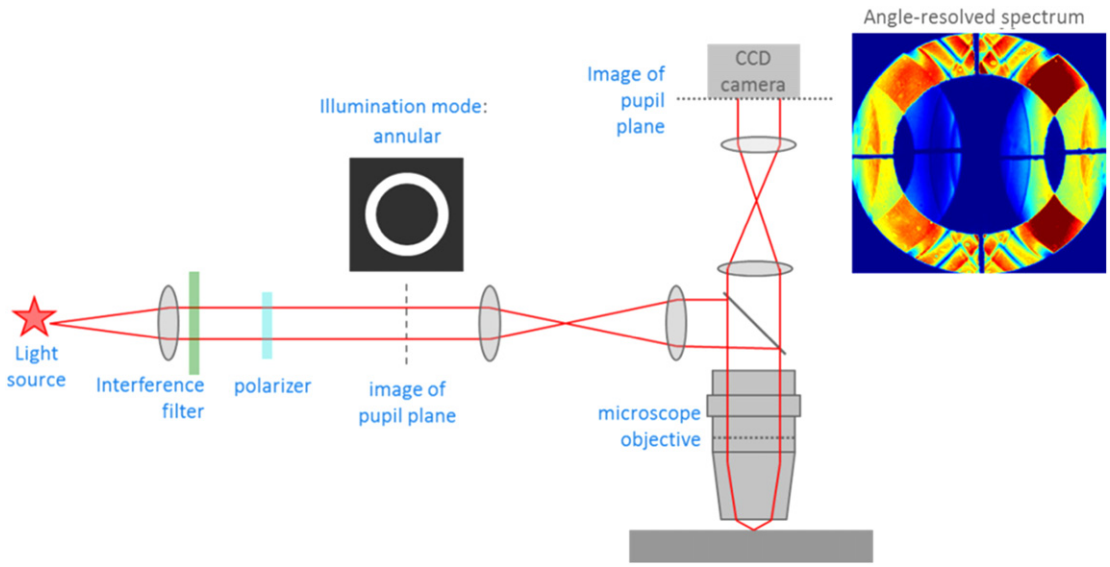


Figure 15. An angle-resolved scatterometer offers various illumination modes for measuring higher diffraction orders. This particular drawing shows how an annular illumination mode is used to measure the +1st and −1st diffraction orders on a DBO overlay target.

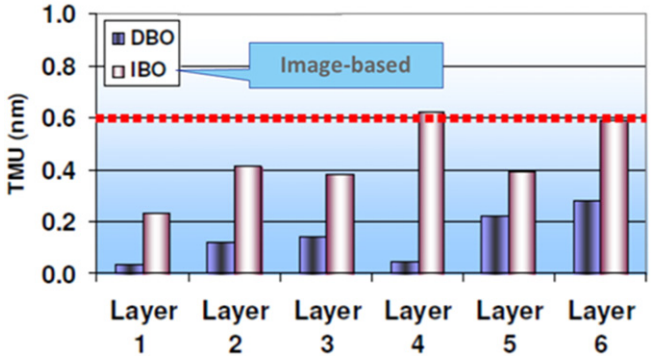


Figure 16. Measured TMU on several product layers. The TMU was measured using IBO and DBO. The red dashed line shows the requirement for this specific production process. The superior performance of DBO is clearly visible.

However, a good TMU is not sufficient for good overlay metrology performance. We want to know the overlay at the device level while we measure on metrology gratings that have a significantly large feature size. This can lead to metrology offsets that are significantly larger than the measured TMU numbers. The main reason for the metrology offsets are:

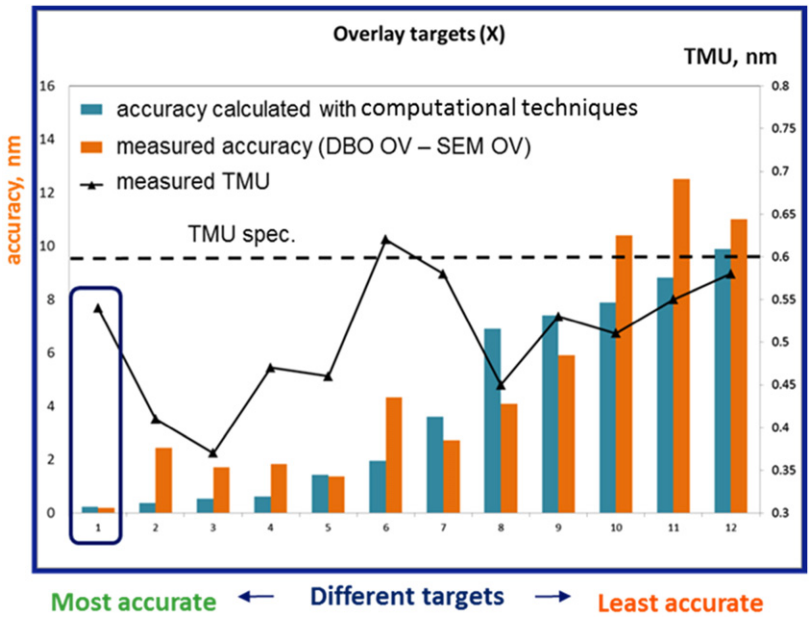


Figure 17. Overlay metrology performance for different grating types. The black solid line shows the measured TMU (right-side vertical axis) and the bars show the measured (brown bar) and simulated (blue bar) accuracy (left-side vertical axis). It can be seen that the best TMU does not yield the highest accuracy.

(1) Processing steps like CMP and etch can cause a small asymmetric distortion of the metrology grating.

(2) Odd aberrations in the projection lens result in a small shift of the metrology target relative to the device pattern which shows up in a metrology offset.

These metrology offsets cannot be solved by better optical sensor concepts since photons contain no information about these errors. Fortunately, very promising results are being obtained today by using context information about the stack in combination with computational metrology grating design techniques. The result of this approach is, for example, a grating where the unit cell has been segmented with device features which results in a lower aberration sensitivity. This computational approach can be extended to the selection of the best possible measurement setting where the sensitivity to asymmetric grating distortion is minimized. These approaches already take us close to the regime of sub-nm accuracy [37] as shown in figure 17.

Figure 17 shows the metrology performance for different target (=grating) types on a specific product wafer. The left axis shows the accuracy performance (matching to the device overlay) and the right axis shows the precision (=TMU) performance. In the accuracy data the overlay as measured with an electron micrscope on the device patterns is used as the reference against which the DBO results are compared. It is interesting to see that a good TMU performance does not yet guarantee a good accuracy. In this particular example, accuracy levels of about 0.5 nm prove to be possible for a target with a sub-optimal TMU performance. The data in figure 17 also shows that there is a good correlation between simulated accuracy and measured accuracy which demonstrates the potential for improving accuracy with computational target design techniques.

Concluding remarks

This paper has presented an overview of various wafer metrology sensors that are needed for good CD and overlay performance. These sensors have in common that they need to offer sub-nm precision on wafers with a large variety of layers. Moreover, they also need to be robust against process variations that are inevitable in a high-volume manufacturing environment. All these sensors rely on the use of gratings either inside the sensor or as metrology targets which ensures a good detectability of the signal over a large range of signal levels. Different solutions are used for robustness against process variations. The level sensor uses UV light to accomplish this. Alignment and Overlay metrology accomplish process robustness by an optimization of the measurement setting (wavelength) and the metrology grating (pitches and process segmentation).

All metrology sensors offer a level of sub-nm precision which is sufficient for future semiconductor manufacturing requirements. The step towards good accuracy is more challenging since this often requires

additional context information that cannot be obtained by direct optical means. We have shown in the section on overlay metrology that accuracy can be significantly improved by computational target design techniques where impact of projection lens aberrations and wafer processing is included in a model that is used in the target design. Very promising results have been obtained and it is expected that these kinds of computational approaches will gain importance in future wafer metrology developments.

References

[1] Moore G E 1965 Cramming more components onto integrated circuits *Electronics* **38** 8

[2] van Zant P 2014 *Microchip Fabrication* (New York: McGraw-Hill)

[3] Bouchoms I, Mulkens J, de Putter S, Gunter P, de Graaf R, Beems M, Verdurmen E, Jasper H, Dieckmanna N and Bornebroek F 2010 Advanced imaging with 1.35 NA immersion systems for volume production *Proc. SPIE* **7640** 76401R

[4] Peeters R *et al* 2014 EUV lithography: NXE platform performance overview *Proc. SPIE* **9048** 90481J

[5] Finders J, Dusa M, Vleeming B, Hepp B, Maenhoudt M, Cheng S and Vandeweyer T 2009 Double patterning lithography for 32 nm: critical dimensions uniformity and overlay control considerations *J. Micro/Nanolith. MEMS MOEMS* **8** 011002

[6] Stout K J and Blunt L 2000 *Three Dimensional Surface Topography* (Amsterdam: Elsevier)

[7] Djordjevic B B and Chern E J 1990 Nonintrusive ultrasonic low-liquid-level sensor *Mater. Eval.* **4** 48

[8] Makosch G and Drollinger B 1984 Surface profile measurement with a scanning differential ac interferometer *Appl. Opt.* **23** 24

[9] van de Werf J E 1992 Optical focus and level sensor for wafer steppers *J. Vac. Sci. Technol.* **10** 2

[10] den Boef A J 2013 Optical metrology of semiconductor wafers in lithography *Proc. SPIE* **8769** 876907

[11] Rubingh R, Moers M, Suddendorf M, Vanoppen P, Kisteman A, Thier M, Blahnik V and Piper E 2005 Lithographic performance of a dual Stage, 0.93NA ArF step&scan system *Proc. SPIE* **5754** 681–92

[12] de Boeij W *et al* 2013 Extending immersion lithography down to 1x nm production nodes *Proc. SPIE* **8683** 86831L

[13] Keij S, Setija I, van der Zouw G and Ebert E 2005 Advances in phase-grating-based waferalignment systems *Proc. SPIE* **5752** 849–60

[14] Nagayama T, Nakajima S, Sugaya A, Kanaya Y and Sukegawa. A 2003 New method to reduce alignment error caused by optical system *Proc. SPIE* **5038**

[15] Magome N, Ota K and Nishi K 1990 New alignment sensors for optical lithography *Japan. J. Appl. Phys* **29** 11

[16] Jacobs B A J and Kramer P 1974 Aparatus for aligning a mask with respect to semiconductor substrate *US* 3,811,779

[17] Bouwhuis G and Wittekoek S 1979 Automatic alignment system for optical projection printing *IEEE Trans. on Electron Devices* **ED-26** pp 723–8

[18] Kreuzer J 2004 Self referencing mark independent alignment sensor *US* 6,809,827 B2

[19] Bunday B, Germer T A, Vartanian V, Cordes A, Cepler A and Settens C 2013 Gaps analysis for CD metrology beyond the 22 nm node *Proc. SPIE* **8681** 86813B

[20] Nyyssonen D 1982 Theory of optical edge detection and imaging of thick layers *J. Opt. Soc. Am.* **72** 1425–36

[21] Diebold A C 2001 *Handbook of Silicon Semiconductor Metrology* (Boca Raton, FL: CRC Press) 0203904540

[22] Habermas A, Hong D, Ross M and Livesay W 2002 193 nm CD shrinkage under SEM: modeling the mechanism *Proc. SPIE* **4689** 92–101

[23] Dixson R, Sullivan N, Schneirl J, McWaidl T, Tsai V W, Prochazka J and Young M 1996 Measurement of a CD and sidewall angle artifact with two dimensional CD AFM metrology *Proc. SPIE* **2725** 572–88

[24] Wang C, Choi K-W, Fu W-E, Ho D L, Jones R L, Soles C, Lin E K, Wu W-L, Clarke J S and Bunday B 2008 CD-SAXS measurements using Laboratory-based and synchrotron-based instruments *Proc. SPIE* **6922** 69222E

[25] Settens C, Bunday B, Thiel B, Kline R J, Sunday D, Wang C, Wu W-L and Matyi R 2013 Critical dimension small angle x-ray scattering measurements of FinFET and 3D memory structures . *Proc. SPIE* **8681** 86810L

[26] Jones R L, Lin E K, Wu W-L, Weigand S J, Keane D T and Quintana J M 2005 Cross sectional measurements of dense high aspect ratio patterns using CD-SAXS *Proc . SPIE* **5752** 404–11

[27] Yang W, Hu J, Lowe-Webb R, Korlahalli R, Shivaprasad D, Sasano H, Liu W and Mui D S L 2004 Line-profile and critical-dimension monitoring using a normal incidence optical CD metrology *IEEE Trans. Semicond. Manuf.* **17** 4

[28] Sendelbach M, Vaid A, Herrera P, Dziura T and Zhang M 2010 Use of multiple azimuthal angles to enable advanced scatterometry applications *Proc. SPIE* **7638** 76381G

[29] Ebert M, Cramer H, Tel W, Kubis M and Megens H 2011 Combined overlay, focus and CD metrology for leading edge lithography *Proc. SPIE* **7973**

[30] Cheung B, Dusa M, Kiers T and Cramer H 2005 Application of optical CD for characterization of 70 nm dense lines *Proc. SPIE* **5752** 30–40

[31] Bhattacharyya K *et al* 2011 New approaches for scatterometry-based metrology for critical distance and overlay measurement and process control *J. Micro/Nanolith. MEMS MOEMS* **10** 1

[32] Ferrieu F *et al* 2007 Metrology and optical characterization of plasma enhanced chemical vapor deposition,(PECVD), low temperature deposited amorphous carbon films *2007 International Conference on Frontiers of Characterization and Metrology* (New York: AIP)

[33] Adel M *et al* 2003 Performance study of new segmented overlay marks for advanced wafer processing *Proc. SPIE* **5038** 453–63

[34] Leray P *et al* 2010 Achieving optimum diffraction based overlay performance *Proc.SPIE* **7638** 76382B

[35] Adel M, Kandel D, Levinski V, Seligson J and Kuniavsky A 2008 Diffraction order control in overlay metrology—a review of the roadmap options *Proc. SPIE* **6922** 692202

[36] Bhattacharyya K *et al* 2015 Holistic approach using accuracy of diffraction-based integrated metrology to improve on-product performance, reduce cycle time and cost at litho *Proc. SPIE* **9424** 94241E

[37] Chen K-H *et al* 2015 Improving on-product performance at litho using integrated diffraction-based metrology and computationally designed device-like targets fit for advanced technologies (incl. FinFET) *Proc. SPIE* **9050** 90500S

[38] Ausschnitt C P 2004 A new approach to pattern metrology *Proc. SPIE* **5375** 51–65